Class - 02

Measure of Central Tendency

Analyzing Data

Measure Of Central Tendency

First thing you'll be thinking "what does this scary name mean?"

Central Tendency simply means **finding the middle point or typical value** of data.

In simple words this means **Finding the middle Point.**

Now this answers the question "What is the center of my data?"

There are several way to describe or find the middle or we can say there are 3 main measures -

1. Mean
2. Median
3. Mode

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1. Mean :-

In simple words, Mean means finding the **Average**.

The definition of mean is -

" Add up all the values, then divide by how many values there are. "

And the Formula is -

$$\text{Mean} = \frac{\text{Number of values}}{\text{Sum of all values}}$$

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Mean :-

Let's see an example of mean,

Suppose we have the Test scores of 5 students

10, 20, 30, 40, 50

$$\text{Mean} = \frac{10+20+30+40+50}{5} = \frac{150}{5} = 30$$

Mean = 30

So, the Mean or Average score is 30.

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2. **Median** :-

Median is also used to find the middle point of our data, but here we are just looking for the middle number.

The definition for Median is -

*" The **middle value** when all data is arranged in order. "*

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Median :-

If the Number of Values or Data is **Odd**,

Median = Middle Value

Let's see an example for Odd values data

Data → 5, 7, 8, 12, 15

Here, Middle Value = **8**

$\therefore$ Median = 8

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Median :-

If the Number of Values or Data is **Even**,

Median = Average of the Two Middle Values

Let's see an example for Even values data,

Data → 5, 7, 8, 12

Here, Middle Value = **7** & **8**

So, Median $= \dfrac{(7+8)}{2} = 7.5$

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3. **Mode** :-

Mode is simple, it just says " **Whichever number occurred most** in our data is the mode. "

The definition for Mode is -

> " The Mode is the number (or category) that **appears most often** in a dataset. "

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Mode :-

There are some Special Cases for Mode -

- **No Mode** $\longrightarrow$ When no number Repeats in the Dataset.

Suppose we have a data

2, 4, 6, 8

In this data all the values occurred only **Once.**

That's why it is a **No Mode Case.**

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Mode :-

- **Bimodal** $\longrightarrow$ When two numbers repeat equally often.

Suppose we have a data

2, 2, 3, 3, 4, 5

Here, both 2 and 3 appear twice.

$\therefore$ Mode = 2 & 3

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Mode :-

- **Multimodal** → When More than two values repeat the most.

Suppose we have a data

2, 2, 3, 3, 4, 4

Here, 2 , 3 and 4 are appearing the most

$\therefore$ Mode = 2 , 3 & 4